

Quality Management

3 — History and evolution of quality management

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A.Y. 2026/2027 • 28 September 2026



Today

One question, asked again and again for six centuries

Who is responsible for quality, and when do you check it?

Every stage in this story answers that differently. Watch two things move:

- **Responsibility** travels from the craftsman, to an inspector, to a department, to the whole company.
- **The moment of checking** travels backwards — from after production, to during it, to before it.

The direction of travel

Each step costs more to set up and less to run. That trade is the whole subject.

KEEP SCORE

At each stage today, ask yourself: *who would be blamed here if a bad product reached a customer?*
The answer changes five times.

The two axes

Everything today moves along these

RESPONSIBILITY widens



CHECKING moves earlier



WHY THEY MOVE TOGETHER

You cannot check earlier without involving more people. Catching a problem at design means the designer, the buyer and the salesperson all have quality duties — not just the man at the end of the line with a gauge.

PART 1

Before the factory

*When the person who made it was the person who
checked it*

The artisan

Producer and controller in one person

Quality first appears with the **artisan** approach to work, where the person who made the object was simultaneously its producer and its controller.

The **mark** affixed to a product was an early indicator of quality: it identified the producer and fixed their responsibility for what they had made.

THE MARK IS STILL WITH US

A hallmark on silver, a maker's stamp on a blade, a brand on a box, the signature on a Declaration of Conformity from last lecture. All the same idea: tie the object to someone who can be held to account for it.

WHAT THIS SYSTEM IS EXTREMELY GOOD AT

Feedback is instantaneous and free. The maker sees every defect, understands its cause, and corrects the next piece — with no meeting, no form and no delay. Nothing we build afterwards ever quite recovers that.

The guilds

The first time the process was written down

With the **guilds** of the Middle Ages, the working methods underlying the master's know-how were formalised for the first time. Written transmission of know-how made two things possible:

- **Repeatability of supplies** — a fundamental idea in quality, and the first time quality outlives one person
- **Preservation of the craft** — the knowledge survives the master

APPRENTICE, JOURNEYMAN, MASTER

A guild qualified *people*, not products, through years of supervised practice ending in a judged piece of work. That is personnel certification — one of the three kinds we saw last lecture — six hundred years early.

BUT NOTICE THE LIMITATION

A guild controlled quality by controlling who was allowed to work. What happens to that model the moment you need two hundred people in one building, most of them trained in weeks?

- **1 Maker and checker were the same person.** Feedback was immediate, free and perfectly informed.
- **2 The mark carried responsibility** — the direct ancestor of today's brand and of the signature on a Declaration of Conformity.
- **3 Guilds wrote the method down**, which made supply repeatable and let knowledge outlive the individual master.
- **4 Quality was controlled by controlling who could work.** That model breaks the moment volume arrives — which is the next section.

PART 2

Industry breaks the link

*Volume rises, the maker stops being the checker, and
someone has to invent inspection*

The first industrial revolution

Great Britain, late 18th century

A strong push towards a far more formalised concept of quality, driven by three things:

New machinery

thermal energy from coal

Rail transport

goods move far from where they are made

Division of labour

no one person makes the whole thing

Quantities produced increased considerably. And with that, quality results depended *less* on the skill of individual operators and *more* on the design and formalisation of the production process.

The break

Once the maker is no longer the checker, quality stops being a personal virtue and becomes an organisational problem. Everything that follows is a response to this.

Interchangeability, and the birth of the tolerance

Where this course's second half comes from

A craftsman fits each part to its neighbour by hand. That works until you need to repair a machine in the field, or assemble from parts made in three different workshops.

Interchangeable parts demand something new: **every part must fall within stated limits**, so that any one of them fits any assembly. That single requirement invents three things at once:

The tolerance

a specified range, not a single number

The gauge

a physical device to judge conformity fast

The inspector

someone whose whole job is to apply it

WHY THIS MATTERS FOR THE EXAM AND FOR THE REST OF THE COURSE

“Conformance to requirements” only becomes a *measurable* idea once requirements are expressed as limits. Everything we do from Lecture 7 onward — statistics, measurement, uncertainty — exists because of this step.

The second industrial revolution

From 1890

Technological innovation and the exploitation of **electrical energy** transformed industry again, and the division of labour became far more pronounced — exacerbated by the introduction of the **Ford-type assembly line**.

The market of those years was characterised by:

low volumes

skilled labour

lack of standardization

WHAT THE MOVING LINE DOES TO QUALITY

On a moving line, a worker has a fixed number of seconds per unit and cannot stop to investigate an oddity. The line converts quality from something the worker *decides* into something the system must *guarantee* — or catch later.

Scientific management

Frederick Taylor, 1911 — the intellectual justification for the split

Taylor proposed separating **planning** from **execution**: engineers determine the one best method, and workers carry it out. Efficiency rose dramatically.

It also had a consequence Taylor did not emphasise: if the worker does not plan the work, the worker is not responsible for the result of the work. **Quality became somebody else's department by design.**

HOLD THIS THOUGHT UNTIL JAPAN

Everything the Japanese quality movement does after 1950 — quality circles, engagement of people, the operator who can stop the line — is an attempt to undo precisely this separation, without giving up the efficiency it bought.

WORTH ARGUING ABOUT

Was Taylor wrong, or just incomplete? Consider that the 1920s alternative was not craft pride — it was unstandardised work of wildly variable quality.

The Testing Department

Before the First World War

In the years before the First World War, organisations began to rely on **inspection and testing**. “Quantity” remained the objective of production, while “quality” was entrusted to a new and separate entity: the **Testing Department**.

What it solved

Bad product stopped reaching customers. For the first time there was a place where quality was somebody’s actual job.

What it created

An adversarial relationship. Production is measured on output, inspection on rejects. The two departments are now structurally in conflict.

A DEFECT FOUND HERE HAS ALREADY COST YOU EVERYTHING

By the time an inspector rejects a finished part, you have paid for the material, the machine time, the labour and the handling. Inspection does not create quality — it sorts it, after all the money is spent.

PART 3

Statistics changes the question

From “is this piece good?” to “is the process behaving?”

Statistics enters

1920 – 1945

Quality in the modern sense of the term begins in the **1920s**, driven by the birth of the first large companies — complex organisations that had to subject process variables to strict control, at ever-lower cost.

Between 1920 and 1945, **statistical quality control** developed. Techniques appeared for controlling the entire production process, rather than only verifying finished products at the end.

Why it had to happen

Blanket checks on every product were becoming too expensive. By examining a *few* finished products, it became possible to establish, *while still producing*, whether the process had irregularities.

THE CONCEPTUAL LEAP

Inspection asks “is this piece good?”. Statistics asks “is the *process* behaving?”. The second question can be answered before the bad pieces exist.

Shewhart and the control chart

Bell Telephone Laboratories, 1924

Walter Shewhart made one observation that reorganised the whole field: **not all variation is the same kind of variation.**

Common-cause variation

The ordinary noise of a stable process. It is the process being itself. Reacting to it individually makes things *worse*.

Special-cause variation

Something changed — a tool, a batch, an operator, a temperature. This is worth hunting down.

The **control chart** is simply a way to tell them apart, in real time, on the shop floor, by someone who is not a statistician.

WHY THE DISTINCTION IS WORTH MONEY

Adjusting a machine because one part drifted slightly — when the process was perfectly stable — injects variation that was not there before. Shewhart's chart tells the operator when *not* to intervene. That is the rarer and more valuable instruction.

- **1 Volume broke the maker/checker link.** Quality became an organisational problem rather than a personal virtue.
- **2 Interchangeability invented the tolerance, the gauge and the inspector** — and with them, measurable conformance.
- **3 Taylor made the split official:** engineers plan, workers execute — so quality became another department's job.
- **4 The Testing Department sorted quality; it did not create it.** Every defect it found had already cost full price.
- **5 Shewhart changed the question** from “is this piece good?” to “is the process behaving?”, separating common-cause from special-cause variation.

PART 4

Quality Control

A filter with no feedback

Quality Control (QC)

1935 – 1945, then spreading through the 1950s

QC was conceived as a simple **pass/fail filter** applied solely to the manufacturing process.

In the 1950s it began to reach other areas of the company — procurement, organisation, design — particularly in the **aeronautical and military** sectors, through:

- predictive calculations of product reliability
- prototype homologation
- control of non-conformities and modifications
- preservation of project and certification documents

But the perspective did not change

QC remained an *intervention*: verify activities or products against specific requirements, and apply a pass/fail filter. It continued to be used **without any feedback mechanism**. Nothing it learned went back upstream.

What 1950s QC looked like in practice

- **Preliminary technical evaluation of suppliers**
- **Constraints on the organization** — minimum requirements for a clear division of responsibilities, especially in quality control activities
- **Qualification of personnel** involved in inspections
- **Qualification of materials**
- **Qualification of certain production processes** identified in advance — the *special processes*
- **Regulation of the management of project documentation**

SPECIAL PROCESSES

Welding, heat treatment, painting — processes whose result you cannot fully verify by inspecting the finished piece. If you cannot inspect the output, you must qualify the process. That principle never went away: it is why welders are certified people and why ISO 9001 has a clause about it.

PART 5

Quality Systems

Japan stops sorting defects and starts preventing them

Quality System (QS)

Post-war Japan

Following the Second World War, Japan pioneered a **systematic approach to quality** as the means to revitalise a shattered economy and reclaim its standing. The new paradigm focused on **process and production**, aiming to produce superior goods at lower cost — rather than simply meeting existing industry standards.

Japanese businesses recognised that **organisational structure and strategy** mattered as much as technical specifications. By implementing robust quality systems, they sought to **prevent defects** and to earn customer trust.

The shift

From **reactive** quality control — find the bad ones — to a **proactive** approach: build a system that does not make bad ones. This is the single most important turn in the whole story.

Why Japan, and why then

The conditions that made it possible

- **Nothing to defend.** Industry had to be rebuilt anyway, so there was no established way of working to protect.
- **No raw materials and no domestic market of scale.** Survival meant exporting — and exporting meant competing on something other than price.
- **A terrible reputation.** “Made in Japan” in 1950 meant cheap and unreliable. That was the problem to be solved.
- **Statistical methods arrived with teachers.** Deming, and later Juran, lectured to Japanese engineers *and to senior management* — which is the part that mattered.

THE DETAIL THAT EXPLAINS EVERYTHING

In the United States, statistical quality control was taught to engineers. In Japan it was taught to **company presidents**. A method that requires investment before it returns anything can only be adopted by someone who controls the budget.

The Quality Assurance Program (QAP)

18 criteria, from 18 CFR 50 Appendix B — aerospace and nuclear

Grouped by what they actually govern — the original lists them as eighteen numbered items.

Govern the organisation

Organization • QAP development • Procedures • Document control
• Quality records • Audits

Govern the work

Operational procedures • Inspections • Testing • Measurement
equipment • Handling, storage and shipping

Govern what comes in

Design control • Purchasing • Supplier control • Identification and
control of materials

Govern what goes wrong

Inspection, testing and operability status • Nonconforming items •
Corrective action

Read the list as an argument

Every criterion exists because something once went wrong without it. In nuclear and aerospace, “we will be careful” is not a control. Notice also that these four groups are almost exactly the clause structure of ISO 9001 — thirty years early.

- **1 QC was a pass/fail filter with no feedback.** It spread beyond manufacturing in the 1950s but never changed its nature.
- **2 Special processes force you to qualify the process** when you cannot inspect the output — a principle still in ISO 9001 today.
- **3 Japan moved from reactive to preventive.** Build a system that does not produce defects, rather than one that finds them.
- **4 It worked partly because it was taught to management,** not only to engineers. Prevention requires spending before returning.
- **5 The 18 QAP criteria are ISO 9001 in embryo** — organisation, inputs, work, and what goes wrong.

PART 6

Total quality

*Quality stops being a department and becomes
everyone's job*

Total Quality Control (TQC)

Feigenbaum, 1950s

Definition

An effective system for integrating the development, maintenance and improvement efforts of quality by the various groups of an organisation, so that marketing, engineering, production and service are carried out at the highest levels of economy, compatible with customer satisfaction.

A global approach is necessary because quality is directly influenced by **nine fundamental factors** — Feigenbaum's *nine Ms*:

Markets • Money • Management • Men • Motivation • Materials • Machines and mechanization • Modern information methods • Mounting product requirements

THE POINT OF THE LIST

Only two of the nine are technical. Quality had become a management problem, and Feigenbaum was the first to say so in those terms.

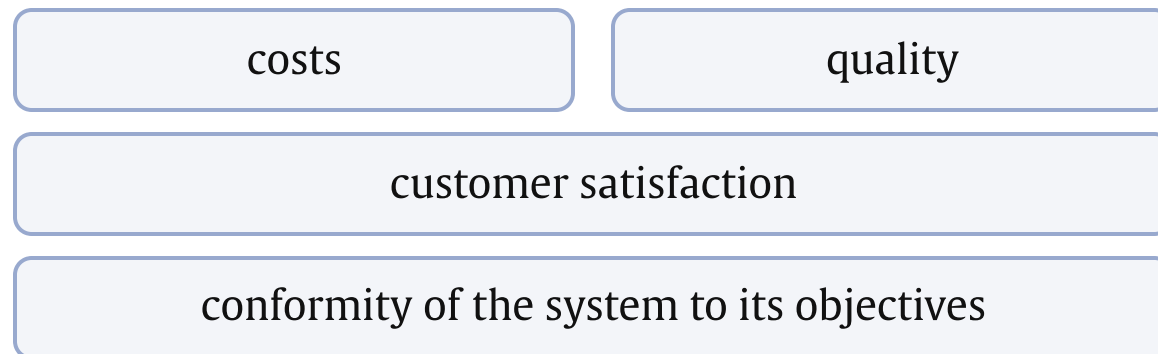
How TQC is put into practice

Sequential phases

- planning the quality of processes and products
- planning, evaluation and control of the quality of *purchased* processes and products
- evaluation and control of the quality of processes and products
- use of quality feedback information
- means of quality information
- orientation, training and development of personnel capabilities
- special studies on quality

What management monitors

Verification actions measure the overall trend of quality in four areas:



Where TQC goes wrong

As with every formalised system

- **Bureaucratic degeneration** — the documentation becomes the point, rather than the quality it was meant to produce.
- **Confusion in relationships** during the introduction phase — nobody is quite sure who decides what any more.
- **A permanent need** for awareness-building and formalisation of personnel — otherwise the tension simply decreases and the system quietly dies.
- **High initial costs** — it is a long-term investment, and it is judged in the short term.

Worth saying out loud

These are not reasons to avoid TQC. They are the reasons most TQC programmes fail, and knowing them in advance is most of the defence.

PREDICT

Which of the four kills most programmes, in your view? Keep your answer — we test it against ISO 9001 in two lectures' time.

Company Wide Quality Control (CWQC)

Deming and Juran take statistics to Japan, 1940s–50s

Deming and Juran exported these concepts and their related statistical tools to Japan, where everything was synthesised into **Company Wide Quality Control**.

CWQC is a quality management *strategy* that aims to involve the entire company structure in improvement. The customer is an essential condition for the company's survival, and the primary objective is their satisfaction, achieved through continuous improvement.

Departmental objective
conformity to specifications

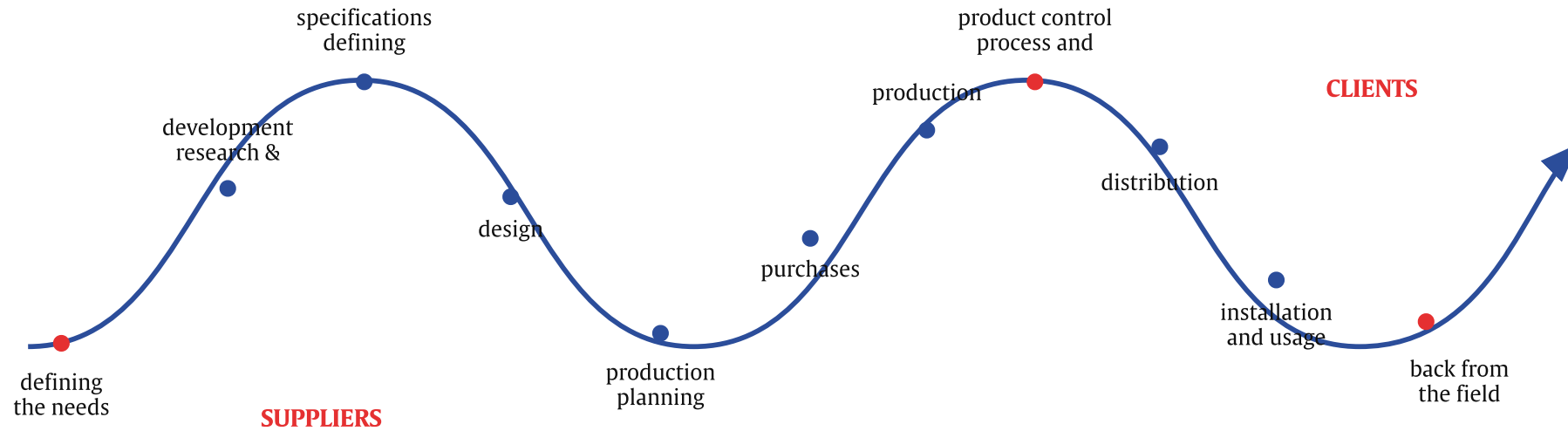
Company objective
fitness for use

THE GAP BETWEEN THOSE TWO BOXES

A product can conform to every specification and still be useless to the customer — because the specification was wrong. *Fitness for use* is judged by the user, not by the drawing. That gap is the subject of the next lecture.

The spiral of quality

The sequence through which fitness for use is achieved



It is a spiral, not a circle: each pass should come back at a higher level than the last.

Seven steps to a quality leap

A universally valid sequence — steps 1 to 3

- **1. Demonstrate the need to intervene** — chronic problems no longer trigger detectable alarm signals. You have to go looking, with studies that identify where intervention is needed.
- **2. Identify improvement projects** — the principle of the “*few essentials, many modest*”. Properties are chosen case by case on: return on investment, urgency, ease of solution, probable resistance to change, expected duration of benefits.
- **3. Organize to guide the projects** — assign responsibility for directing each project’s progress and for bringing the necessary programmes to completion.

“FEW ESSENTIALS, MANY MODEST”

This is the Pareto principle in Juran’s own words. A small number of causes produce most of the loss — find them before spending anything. We build the Pareto chart itself in the statistics lectures.

Seven steps to a quality leap

Steps 4 to 7

- **4. Diagnose** — make the journey from symptoms to causes, and from causes to remedies: understand the symptoms, form hypotheses about causes, choose which to verify, test them, and design a remedy — usually a technological or a management change.
- **5. Overcome cultural resistance** — resistance is rarely about the technology. It is social: habits and status of the people involved, who perceive the change as a threat to their cultural model.
- **6. Iterate the corrective actions** — once a remedy is applied, do not stop. Move forward until all the possibilities the change opened up are fully exploited.
- **7. Control at the new level** — consolidate what was achieved, and make it the starting point for the next improvement.

STEP 5 IS THE ONE ENGINEERS UNDERESTIMATE

A technically correct solution that nobody adopts has solved nothing. Juran put cultural resistance in the middle of the sequence, not in an appendix, because that is where projects actually die.

Crosby, and where our definition came from

1979 — the argument aimed at management, not engineers

Philip Crosby made three claims that are still the most quotable in the field:

- **Quality is conformance to requirements** — the definition we adopted in Lecture 1, chosen deliberately to stop arguments about taste.
- **The performance standard is zero defects** — not as a slogan, but as a rejection of the idea that an “acceptable” defect rate should be planned for.
- **Quality is free** — what costs money is *non*-quality. Prevention is paid for by the failures it avoids.

WHY “QUALITY IS FREE” WAS AIMED AT FINANCE, NOT ENGINEERING

Engineers already believed in quality. The people who had to be convinced controlled the budget, and they heard “quality” as “cost”. Crosby’s contribution was to reframe it as a saving — which is exactly the calculation we do next lecture.

The cast, in one slide

Who contributed what — worth knowing by name

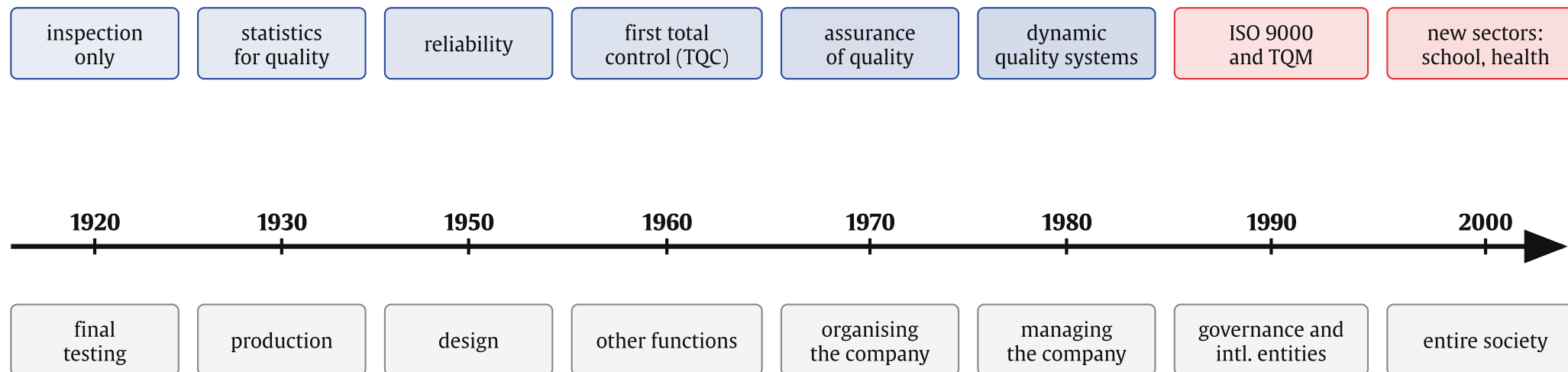
- **Shewhart** — the control chart; common-cause versus special-cause variation. The statistical foundation everything else stands on.
- **Deming** — carried statistical methods to Japan and taught them to management; the improvement cycle we know as PDCA.
- **Juran** — fitness for use; the quality spiral; the “few essentials”; the seven-step quality leap.
- **Feigenbaum** — Total Quality Control and the nine Ms; quality as a management problem.
- **Ishikawa** — the cause-and-effect diagram, and quality circles: the tools that let operators participate.
- **Crosby** — conformance to requirements; zero defects; quality is free.

A FAIR EXAM QUESTION

Given a concept from this course, name the person — and, harder, say what problem they were reacting *against*.

The century in one picture

Above: how quality was done • Below: who was involved



Both rows move the same way: outward from the workshop floor, and backwards in time.

Where we got to **TAKEAWAYS**

Six centuries in six lines — 1 of 2

- **1 Artisan** — maker and checker are the same person; the mark carries the responsibility.
- **2 Industry** — the link breaks; interchangeability brings tolerances and gauges; a separate Testing Department appears, and Taylor makes the split doctrine.
- **3 1920s** — Shewhart moves the question from the piece to the process, and separates the variation worth chasing from the variation you should leave alone.

Where we got to **TAKEAWAYS**

Six centuries in six lines — 2 of 2

- **1 QC** — a pass/fail filter, with no feedback upstream.
- **2 Quality systems** — post-war Japan turns from reactive to preventive, and teaches it to management.
- **3 TQC / CWQC** — quality becomes the whole company's job; the objective moves from conformity to specifications to *fitness for use*.

Next time

What quality actually *is*, now that it is defined by the user — and what it costs. The costs of non-quality run at roughly 25–40% of total costs.

Questions?

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